

Skills and Jobs Academy

Web Application Development with Python Level-4

Written Demo Assessment-01

Time: 50minute

Mark: 30

Name:

Part 1: Multiple Choice Questions

1. Which of the following is NOT a web server commonly used with Django?

- A. Apache
- B. MySQL
- C. Nginx
- D. IIS

2. Which command is used to create a new Django project?

- A. django new project_name
- B. django-admin startproject project_name
- C. create-django-project project_name
- D. python manage.py startproject project_name

3. In which file should you define the URL patterns for your Django application?

- A. settings.py
- B. urls.py
- C. models.py
- D. views.py

4. In a typical Django project, what is the primary configuration file that contains settings for the project?

- A. settings.py
- B. urls.py
- C. models.py
- D. views.py

5. What is the primary function of the views.py file in a Django project?

- A. It defines the database schema and models.
- B. It specifies the URL patterns and their corresponding views.
- C. It contains the project's settings.
- D. It contains the logic for handling HTTP requests and rendering templates.

6. How can you capture a variable value from a URL in Django?

- A. Using the @ symbol in the URL
- B. Using regular expression in the URL pattern
- C. Using square brackets in the URL
- D. Using double curly braces in the URL pattern

7. Which filter is used to join elements of a list into a single string in Django templates?

- A. merge
- B. join
- C. concat
- D. combine

8. Which command is used to create the first project in Django?

- A. `django-admin startproject project_name`
- B. `django-admin startapp project_name`
- C. `django-admin create project_name`
- D. `django-admin createproject project name`

9. What is the purpose of the 'related_name' attribute in a Django model's foreign key field?

- A. It specifies the name of the related model
- B. It provides a way to access related objects from the related model
- C. It specifies the database table name for the related model
- D. It defines the default value for the foreign key field

10. In Django model managers, what is the purpose of the 'create()' method?

- A. Updates an existing record
- B. Creates a new record in the database
- C. Deletes records based on a filter
- D. Retrieves a single record from the database

11. Which method is commonly used in Django views to handle the Delete operation (removing data) for a specific model instance?

- A. `remove()`
- B. `erase()`
- C. `destroy()`
- D. `delete()`

12. What is the purpose of 'DateTimeField' in Django models?

- A. Stores only date information
- B. Stores only time information
- C. Stores both date and time information
- D. Stores duration or time interval

13. Which template tag is used to display the URL for a named URL pattern in Django templates?

- A. `{% url %}`
- B. `{% link %}`
- C. `{% href %}`
- D. `{% route %}`

14. How do you access dictionary values in Django templates?

- A. `{{ dictionary.key }}`
- B. `{{ dictionary[key] }}`
- C. `{{ dictionary.value }}`
- D. `{{ dictionary.item }}`

15. Which template tag is used for including template comments that won't be rendered in the final HTML output?

- A. {% comment %}
- B. {# comment #}
- C. <!-- comment -->
- D. {% ignore %}

16. How do you create a dynamic webpage using Django templates?

- A. By writing JavaScript code
- B. By defining dynamic content using template tags and filters
- C. By using HTML only without any dynamic elements
- D. By including external PHP files

17. What is the purpose of the python manage.py migrate command in Django?

- A. It applies all the pending database migrations to the database
- B. It creates new database migrations based on the changes detected in the models.
- C. It initializes a new Django project with a superuser account.
- D. It generates a new migration file for the database schema.

18. Which of the following statements is true about virtual environments in Python?

- A. Virtual environments are used to run Python code without installing Python globally on the system.
- B. Virtual environments are used to create separate instances of the operating system.
- C. Virtual environments are not necessary in Python as it supports global package installations.
- D. Virtual environments can only be created using third-party packages, not with the built-in Python tools.

19. What is the purpose of the pip freeze command in Python?

- A. It installs Python packages.
- B. It lists all installed packages and their versions.
- C. It upgrades the Python interpreter.
- D. It uninstalls Python packages.

20. Which of the following is used to define a primary key in a Django model?

- A. PrimaryKey
- B. primary_key=True
- C. primary=True
- D. pk=True

Part 2: Short Answer Questions

21. What is the purpose of the migrations directory in a Django project?

22. What is the significance of the 'name' parameter in a URL pattern?

23. What is the advantage of using named URL patterns in Django?

24. What does the yesno filter do in Django templates?

25. What is the purpose of the save() method in Django models?

26. What is the role of the 'TemplateView' class in Django CBVs?

27. What is the significance of the 'request.POST' dictionary in Django form submissions?

28. What challenges can arise when managing versioning and caching for static files in large web applications?

29. What are Bootstrap cards, and how are they used in web development?

30. How can you authenticate users in DRF APIs?